

Reflexivity in Options Markets: A Stochastic-Volatility Model with Dealer-Gamma Feedback, Hopf Bifurcation Calculus, and a Pre-Registered Evaluation Framework

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Abstract

We study a continuous-time Heston-like stochastic-volatility model in which the underlying spot price is coupled to its own option market through a dealer-gamma feedback channel, derived from the open-interest grid via a Gârleanu–Pedersen–Poteshman [19] demand-pressure mapping. The model state is three-dimensional—spot S_t , instantaneous variance v_t , and a low-pass-filtered log-spot memory variable z_t —and reduces to standard time-dependent Heston [25] at zero coupling. We synthesise three previously separate ingredients: (i) a continuous-time SV backbone with explicit dealer-gamma drift; (ii) a formal Hopf bifurcation calculus on the deterministic skeleton with first Lyapunov coefficient ℓ_1 and Khasminskii-style stochastic shift Λ ; and (iii) a pre-registered sliced-Wasserstein-2 evaluation protocol for RL-trained derivatives agents. For a representative dimensionless calibration we obtain $\kappa^* \approx 0.8964$, $\omega^* \approx 0.5724$ rad/yr, and $\ell_1 \approx -0.025 < 0$ (supercritical: an attracting limit cycle in volatility is born for $\kappa > \kappa^*$). The 2D reduction has an upper-triangular Jacobian and admits only saddle-node onset, recovering the discrete-time recursion of Dai [15]; the 3D extension is what unlocks the Hopf, distinguishing this work from Halperin and Itkin [21]’s Marketron quasi-particle SDE. For log-normal open-interest in moneyness we obtain a closed-form Hopf threshold and a closed-form ℓ_1 , exposing a parametric phase boundary in (σ_q, γ) space. We additionally close three theoretical loose ends: (i) a codim-2 Bautin/Bogdanov–Takens analysis showing the BT locus is empty in the canonical scan window (Theorem 2); (ii) the McKean–Vlasov mean-field correction to the single-dealer threshold, regime-dependent in sign ($\kappa_{\text{MV}}^*/\kappa_{\text{single}}^* < 1$ strictly at the canonical short-gamma regime, with frozen-dealer limit $8/21$ and first-order asymptote $1 - 0.664/\theta_G + O(\theta_G^{-2})$; Theorem 3); and (iii) a Hawkes–SV equivalence at the Hopf boundary identifying the empirical Hardiman et al. [22] branching ratio $n \approx 1$ with $\kappa = \kappa^*$ via the Bacry et al. [3] diffusive-limit identity (Theorem 4).

Keywords

Reflexivity, dealer gamma, Hopf bifurcation, stochastic volatility, pre-registration, sliced Wasserstein, RL for finance

1 Introduction

Real options markets exhibit endogenous volatility cycles that pure stochastic-volatility (SV) models cannot reproduce. The empirical anchor is Hardiman et al. [22], who fit Hawkes-process branching ratios to E-mini mid-price changes over 1998–2011 and find $n \approx 1$ throughout—the precise critical edge between sub- and super-criticality of endogenous events. Soros [42] and Bouchaud [10] are the conceptual antecedents; Filimonov and Sornette [17] formalised the reflexivity-quantification programme that Hardiman et al. [22] executed. We ask what continuous-time mechanism produces this

critical reflexivity, and how the analytical structure that quantifies it should be evaluated against historical surface dynamics.

We contribute the following.

- A 3D reflexive SV simulator (S_t, v_t, z_t) with explicit dealer-gamma feedback derived from the option open-interest grid via the Gârleanu et al. [19] demand-pressure mapping. The variance backbone is Heston [25]; the memory equation $\dot{z} = -\alpha z + \beta \log(S/S_0)$ closes the leverage feedback channel that the 2D reduction lacks.
- A Hopf bifurcation theorem (Theorem 1) for the deterministic skeleton at κ^* given by Liu’s criterion, with closed-form first Lyapunov coefficient ℓ_1 and closed-form Hopf threshold for log-normal open-interest in moneyness (Eqs. 9–10).
- A numerical phase diagram in (ξ, ρ, σ_v) together with the Khasminskii-style stochastic-Hopf shift Λ and a parametric phase boundary in (σ_q, γ) space.
- A pre-registration of the empirical evaluation pipeline (sliced- W_2 over 21-day surface windows, κ -sensitivity slope, Hopf-frequency spectral test) at a verifiable git commit hash with an OpenTimestamps Bitcoin-anchored proof, in the spirit of Camerer et al. [12] adapted for computational finance.

Closest precedents. Halperin and Itkin [21] (Marketron) share the model-structural axis but do not perform the bifurcation analysis. Dai [15] share the dealer-gamma axis but produce a saddle-node onset. He et al. [23] (NeurIPS 2025) share the RL-with-environmental-sweep idea but train separate agents per grid point rather than reporting the slope at an anchor. The continuous-time HAM-Hopf precedents [14, 24] use delay-differential heterogeneous-belief models without an options-grid feedback channel; Brock et al. [11]’s “more hedging instruments may destabilize markets” result is the methodological template for derivative-driven Neimark–Sacker bifurcations. The hedger-flow-feedback foundations are Frey and Stremme [18], Platen and Schweizer [37], Sircar and Papanicolaou [41].

2 The reflexive model

Let S_t be the underlying spot price, v_t the instantaneous variance, and z_t a memory variable encoding low-pass-filtered price history. The reflexive system is

$$\frac{dS_t}{S_t} = (\mu + \kappa G(S_t, z_t, v_t)) dt + \sigma(S_t, v_t) dW_t^S, \quad (1)$$

$$dv_t = (\kappa_v(\theta_v - v_t) + \gamma z_t) dt + \xi \sqrt{v_t} dW_t^v, \quad (2)$$

$$dz_t = (-\alpha z_t + \beta \log(S_t/S_0)) dt, \quad (3)$$

with $d\langle W^S, W^v \rangle = \rho dt$, reflexive coupling $\kappa \geq 0$, leverage strength $\gamma \geq 0$, memory decay $\alpha > 0$, β dimensionless, and G the aggregate

dealer-gamma exposure derived from the OI grid:

$$G(S, t) = \sum_{K,T} q_{K,T} \Gamma_{K,T}(S, t) \text{sign}(K, T), \quad (4)$$

following the SqueezeMetrics dealer-net convention [43]. When $\kappa = \gamma = 0$ the system collapses to standard Heston [25] (verified numerically). For variance we use the full-truncation Euler scheme of Lord et al. [32].

Why three states. A 2D specification (drop z_t and the γz term) has a Jacobian $J_{2D} = \begin{pmatrix} a(\kappa) & b(\kappa) \\ 0 & -\kappa_v \end{pmatrix}$, upper triangular with real eigenvalues: only saddle-node / transcritical bifurcations are admissible. To Hopf, the feedback must flow in both directions; the leverage channel γz together with the auxiliary memory equation (3) is the minimal structural modification that supplies it—structurally analogous to the memory variable in Marketron [21] though with very different mechanistic content.

3 Hopf bifurcation theorem and closed-form ℓ_1

3.1 Equilibria, Jacobian, Liu criterion

Drop the Brownian terms and work in log-deviation variables $(y, u, z) = (\log(S/S^*), v - \theta_v, z - z^*)$ around the equilibrium (S^*, θ_v, z^*) defined by

$$\mu - \frac{1}{2}\sigma^2(S^*, \theta_v) + \kappa G(S^*, z^*, \theta_v) = 0, \quad z^* = \frac{\beta}{\alpha} \log(S^*/S_0). \quad (5)$$

The deterministic skeleton becomes $\dot{x} = J(\kappa)x$ with

$$J(\kappa) = \begin{pmatrix} a(\kappa) & b(\kappa) & \kappa G_z \\ 0 & -\kappa_v & \gamma \\ \beta & 0 & -\alpha \end{pmatrix}, \quad (6)$$

where $a(\kappa) = \kappa G_x - \frac{1}{2}\partial_x \sigma^2$, $b(\kappa) = \kappa G_v - \frac{1}{2}\partial_v \sigma^2$. The characteristic polynomial is $P(\lambda; \kappa) = \lambda^3 + c_2\lambda^2 + c_1\lambda + c_0$ with $c_2 = -a + \kappa_v + \alpha$, $c_1 = -a\kappa_v - \alpha\alpha + \kappa_v\alpha - \kappa G_z\beta$, $c_0 = -a\kappa_v\alpha - \beta(\kappa G_z\kappa_v + b\gamma)$. A 3D system Hopfs iff (Liu's criterion):

$$c_2 > 0, \quad c_0 > 0, \quad H(\kappa) := c_1c_2 - c_0 = 0, \quad H'(\kappa) \neq 0. \quad (7)$$

The Hopf threshold is $\kappa^* = \{\kappa > 0 : H(\kappa) = 0, H'(\kappa) \neq 0, c_2, c_0 > 0\}$. At κ^* , J has imaginary pair $\lambda_{\pm} = \pm i\omega^* = \pm i\sqrt{c_1(\kappa^*)}$ plus real eigenvalue $\lambda_3 = -c_2(\kappa^*) < 0$.

3.2 Theorem 1

THEOREM 1 (HOPF BIFURCATION IN THE GAMMA-COUPLED SV SKELETON). *Let the deterministic system (1–3) satisfy: (A1) $G, \sigma^2 \in C^3$ with σ^2 bounded below by a positive constant; (A2) a locally unique equilibrium for every $\kappa \in [0, \kappa_{\max}]$ with non-degenerate linearisation (multiple equilibria may exist globally; analysis is local); (A3) $\kappa \mapsto (a(\kappa), G_z, G_v)$ in C^3 ; (A4) there exists $\kappa^* \in (0, \kappa_{\max})$ satisfying (7) with $\omega^* := \sqrt{c_1(\kappa^*)} > 0$; (A5) $\ell_1(\kappa^*) \neq 0$ [29, eq. 3.20]. Then the equilibrium undergoes a Hopf bifurcation at $\kappa = \kappa^*$: (1) for κ in a one-sided neighbourhood of κ^* there is a unique smooth family of periodic orbits Γ_{κ} of period $T_{\kappa} = 2\pi/\omega^* + O(\kappa - \kappa^*)$ and amplitude $\propto \sqrt{|\kappa - \kappa^*|}$; (2) orbits are stable iff $\ell_1(\kappa^*) < 0$ (supercritical); (3) on Γ_{κ} , $u_t = v_t - \theta_v$ oscillates with amplitude $\propto \sqrt{|\kappa - \kappa^*|}$, providing endogenous volatility cycles.*

PROOF SKETCH. The eigenvalue pair $\lambda_{\pm}(\kappa)$ is smooth near κ^* by the implicit function theorem on P . Routh–Hurwitz gives $\text{Re } \lambda_{\pm}(\kappa^*) =$

0, and $H'(\kappa^*) \neq 0$ translates to $\frac{d}{d\kappa} \text{Re } \lambda_{\pm} \neq 0$. The centre manifold theorem [29, Thm. 5.4] gives a 2D invariant manifold tangent to the eigenspace of $\pm i\omega^*$; conjugation to Poincaré normal form and Kuznetsov [29, Thm. 3.3] yield (1)–(3). $\square$

Numerical anchor. For $\{G_x, G_v, G_z\} = \{0.5, -0.5, -0.5\}$, $\alpha = 0.5/\text{yr}$, $\beta = 1$, $\gamma = 0.5$, $\kappa_v = 2/\text{yr}$ (constant-vol surrogate): $\kappa^* = 0.8964$, $\omega^* = 0.5724 \text{ rad/yr}$, $\lambda_3 = -2.052$, $\ell_1 = -2.53 \times 10^{-2}$ (supercritical). Integrating the deterministic skeleton at $\kappa = 1.05\kappa^*$ from random initial conditions, the trajectory converges to a closed orbit whose measured period $T_{\text{measured}} = 10.561 \text{ yr}$ matches the linear prediction $2\pi/\omega^* = 10.977 \text{ yr}$ to 3.79%—within the leading-order $O(\kappa - \kappa^*)$ Hopf-normal-form correction.

3.3 Closed-form κ^* and ℓ_1 for log-normal OI

For log-normal open-interest in log-strike, $q(\log K) \propto \exp(-(\log K - \mu_q)^2 / (2\sigma_q^2))$, the Gaussian-product identity collapses the OI integral to

$$G(a, v) = C(v) \cdot e^{-a} \cdot \exp\left(-\frac{(a - m(v))^2}{2\tau^2(v)}\right), \quad (8)$$

with $C(v) = \kappa_u e^{-q_{\text{div}} T_{\text{eff}}} / \sqrt{2\pi\tau^2(v)}$, $\tau^2(v) = \sigma_q^2 + v T_{\text{eff}}$, $m(v) = \mu_q - (r - q_{\text{div}} + \frac{1}{2}v)T_{\text{eff}}$. With $G_z = 0$ and $\sigma^2 = v$, $H(\kappa) = c_1c_2 - c_0$ degree-collapses from cubic to quadratic; writing $A = \alpha + \kappa_v$, $M = \alpha\kappa_v$, $L = \beta\gamma$, the Hopf threshold is

$$\kappa^* = \frac{G_y A^2 - G_v L - \sqrt{D}}{2 G_y^2 A}, \quad D := (G_v L - G_y A^2)^2 - 4 G_y^2 A (M A - \frac{L}{2}). \quad (9)$$

The discriminant $D < 0$ implies no Hopf at any κ and unconditional stability of the deterministic skeleton.

The third-order partials of (8) are explicit rational functions of $(\delta := a^* - m(v^*), \tau^2, T_{\text{eff}})$, so the bilinear and trilinear tensors B, C at κ^* admit a closed form. Substituting into the Kuznetsov formula

$$\ell_1(\kappa^*) = \frac{1}{2\omega^*} \text{Re} \left[\langle p, C(q, q, \bar{q}) \rangle - 2 \langle p, B(q, J^{-1} B(q, \bar{q})) \rangle + \langle p, B(\bar{q}, (2i\omega^* I - J)^{-1} B(q, q)) \rangle \right], \quad (10)$$

with $Jq = i\omega^* q$, $J^T p = -i\omega^* p$, $\langle p, q \rangle = 1$, yields a ~ 30 -term closed-form rational in the G -partials and $(\kappa_v, \alpha, \beta, \gamma)$, evaluated in $\sim 50 \mu\text{s}$ per parameter tuple and matching the FD-tensor pipeline to $< 0.6\%$. At the canonical regime ($\sigma_q = 0.10$, $T_{\text{eff}} = 0.25$, $\kappa_v = 2$, $\alpha = 0.05$, $\beta = 1$, $\gamma = 1$, $\mu_q = \log 100$, $v^* = 0.04$): $\kappa^* = 17.81$, $\omega^* = 1.18 \text{ rad/yr}$, $\ell_1 = -4.81 \times 10^{-1}$ (supercritical). With the closed form in hand, the sign of ℓ_1 near the boundary contour is structurally certain rather than statistically inferred.

3.4 Calibration tolerance from κ^* sensitivity

Differentiating (9) analytically through the G -partials at the canonical specification ($\kappa^* = 17.81$, $\omega^* = 1.18 \text{ rad/yr}$) yields the elasticities $\eta_{\sigma_q} = -1.583$ (moderate) and $\eta_{\mu_q} = +702.83$ (extreme). μ_q is the binding constraint: a $\Delta\mu_q = 0.001$ shift moves κ^* by $\sim 15\%$, and $\pm 30\%$ in σ_q exits the supercritical Hopf region entirely. For non-log-normal (bimodal-mixture) OI the closed form is essentially exact at $\Delta = 0.05$ (0.23% relative error), acceptable at $\Delta = 0.10$ (4.8%), and fragile at $\Delta = 0.20$ (119%). The Phase-4 calibration tolerances are therefore σ_q to $\pm 2.23\%$ relative and μ_q to within $\sim 5 \text{ bp}$ of ATM in

log-strike to deliver $\pm 5\%$ on κ^* —achievable with a 5-strike-wide ladder and the binding observational requirement.

3.5 Codim-2 bifurcation structure: Bautin and Bogdanov–Takens

The codim-1 Hopf result treats κ^* at fixed (σ_q, γ) . At the boundary of the Hopf region in (σ_q, γ) two codim-2 phenomena are admissible [29, Ch. 8]. *Bautin* (degenerate Hopf) is the locus $\ell_1(\sigma_q, \gamma) = 0$; crossing it flips supercritical (smooth limit cycle) to sub-critical (hysteresis with abrupt jumps). The closed-form ℓ_1 admits row-wise sign-change interpolation on a 71×97 grid over $(\sigma_q, \gamma) \in [0.05, 0.40] \times [0.20, 5.00]$; the sub-critical region dominates by $\sim 5:1$ in cell count. *Bogdanov–Takens* (BT) requires the saddle-node curve to coalesce with the Hopf curve. The constant Routh–Hurwitz coefficient $c_0(\kappa) = -\kappa(G_y \alpha \kappa_v + G_v \beta \gamma) + \frac{1}{2} \beta \gamma$ is linear in κ , so the saddle-node coupling is $\kappa_{\text{SN}}(\sigma_q, \gamma) = \frac{1}{2} \beta \gamma / (G_y \alpha \kappa_v + G_v \beta \gamma)$.

THEOREM 2 (BT LOCUS EMPTY IN CANONICAL SCAN WINDOW). *On the 71×97 grid over $(\sigma_q, \gamma) \in [0.05, 0.40] \times [0.20, 5.00]$ at the canonical specification, $\kappa_{\text{SN}}(\sigma_q, \gamma) \leq -1.31$ at every cell (range $[-68.05, -1.31]$); the Bogdanov–Takens locus is therefore empty in this scan window.*

PROOF SKETCH. $G_v(\sigma_q) < 0$ throughout the scanned range and $|G_v \beta \gamma| > |G_y \alpha \kappa_v|$ (dominance ratio ≥ 1.44 at every cell, reaching ~ 358 at the small- σ_q corner), so the denominator of κ_{SN} is uniformly negative while the numerator $\frac{1}{2} \beta \gamma$ is positive; the quotient is uniformly negative. The unbounded-domain question is left open. $\square$

The economic content is that the dealer-gamma + leverage regime is structurally Hopf-only *within the scanned window*: the model does not admit excitable spike-and-recovery dynamics from a homoclinic emanation. Burst-relax phenomena observed in real volatility surfaces near macro events must therefore originate from non-stationary parameter drift, regime-switching of the OI distribution, or higher-order degeneracies—not from the deterministic skeleton itself.

3.6 McKean–Vlasov mean-field limit ($n_{\text{dealers}} \rightarrow \infty$)

Section 2 treats the aggregate dealer-gamma exposure G as if it came from a single representative market-maker. Real markets have $n \sim 10^2$ – 10^3 dealers with idiosyncratic noise. With each dealer i holding gamma exposure G_i evolving via $dG_i = -\theta_G(G_i - \bar{G}_n)dt + \sigma_G dW_G^i$ and $\bar{G}_n = (1/n) \sum_j G_j$, propagation of chaos gives $\bar{G}_n \rightarrow \bar{G}_\infty$ [45, Ch. 1] with $\sup_{t \leq T} \mathbb{E}[(\bar{G}_n - \bar{G}_\infty)^2] \leq C(T)/n$, $C(T) = \max(\text{Var}(G^0), \sigma_G^2/(2\theta_G))$; cf. Carmona and Delarue [13, Ch. 1] for the modern MV formalism.

4D extended Jacobian. The linearised MV system acquires a fourth state $g(t) := \bar{G}_\infty(t) - G^*$ with bandwidth θ_G ; the spot equation’s instantaneous feedback $\kappa(G_y y + G_v u + G_z z)$ is REPLACED

by κg :

$$J_{\text{MV}}(\kappa, \theta_G) = \begin{pmatrix} -\frac{1}{2}\sigma_y^2 & -\frac{1}{2}\sigma_v^2 & 0 & \kappa \\ 0 & -\kappa_v & \gamma & 0 \\ \beta & 0 & -\alpha & 0 \\ \theta_G G_y & \theta_G G_v & \theta_G G_z & -\theta_G \end{pmatrix}. \quad (11)$$

A Hopf bifurcation occurs at the smallest $\kappa > 0$ satisfying the Liu [31] 4D condition $H_4(\kappa, \theta_G) := (a_3 a_2 - a_1) a_1 - a_3^2 a_0 = 0$ with positivity side conditions $a_3, a_0, a_3 a_2 - a_1 > 0$, where a_i are the coefficients of $\det(\lambda I - J_{\text{MV}}) = \lambda^4 + a_3 \lambda^3 + a_2 \lambda^2 + a_1 \lambda + a_0$.

THEOREM 3 (McKEAN–VLASOV HOPF THRESHOLD, CORRECTED). *Let $\kappa_{\text{MV}}^*(\theta_G) := \inf\{\kappa > 0 : \max_i \text{Re } \lambda_i(J_{\text{MV}}(\kappa, \theta_G)) = 0\}$ with the 4D positivity side conditions. Then:*

- (1) (Instantaneous-hedging recovery.) $\lim_{\theta_G \rightarrow \infty} \kappa_{\text{MV}}^*(\theta_G) = \kappa_{\text{single}}^*$.
- (2) (Regime-dependent sign.) At the canonical short-gamma regime $(G_y, G_v, G_z, \alpha, \beta, \gamma, \kappa_v) = (\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, 1, \frac{1}{2}, 2)$ with $\sigma_y^2 = \sigma_v^2 = 0$,

$$\kappa_{\text{MV}}^*(\theta_G) = \frac{50\theta_G^2 + 143\theta_G + 105 - (2\theta_G + 5)\sqrt{385\theta_G^2 + 810\theta_G + 441}}{12\theta_G(\theta_G + 1)}, \quad (12)$$

with $\kappa_{\text{single}}^* = (25 - \sqrt{385})/6 \approx 0.8964$, and the ratio $\rho(\theta_G) := \kappa_{\text{MV}}^*/\kappa_{\text{single}}^* < 1$ strictly for every finite $\theta_G > 0$.

- (3) (Frozen-dealer limit.) $\lim_{\theta_G \rightarrow 0^+} \kappa_{\text{MV}}^*(\theta_G) = 8/21$, giving $\rho(0^+) \approx 0.425$ (finite).
- (4) (Asymptotic expansion.) $\rho(\theta_G) = 1 - 0.6643 \dots / \theta_G + O(\theta_G^{-2})$ — leading-order correction is first-order in τ_G and NEGATIVE.

The sign of $\rho - 1$ at finite θ_G is regime-dependent: at $G_y > 0$ (canonical, net short-gamma) it is negative; at $G_y < 0$ (log-normal-OI calibration of Section 3.3) it is positive.

PROOF SKETCH. (i) The reducibility of J_{MV} under $\theta_G \rightarrow \infty$ sends the fourth eigenvalue to $-\infty$ and leaves the remaining three to inherit the 3D single-dealer Hopf condition exactly. (ii) Substituting the canonical numerical values into $H_4(\kappa, \theta_G) = 0$ and factoring out θ_G yields the quadratic $6\theta_G(\theta_G + 1)\kappa^2 - (50\theta_G^2 + 143\theta_G + 105)\kappa + 5(\theta_G + 2)(2\theta_G + 1) = 0$, whose smaller positive root is (12); positivity is verified analytically at this root for all $\theta_G > 0$. (iii) Direct substitution. (iv) Taylor expansion in $1/\theta_G$. $\square$

Numerical anchors. Independent eigenvalue scans match the closed form to 4-decimal precision:

θ_G	κ_{MV}^* (scan)	closed form	$\rho(\theta_G)$
0.5	0.5359	0.5359	0.598
1.0	0.6195	0.6195	0.691
5.0	0.7996	0.7996	0.892
50	0.8848	0.8848	0.987
500	0.8952	0.8952	0.999

Table 1: Audit-anchor table for Theorem 3 at the canonical short-gamma regime. Closed-form and numerical eigenvalue scan agree exactly. A 64-replicate n -particle Euler simulation across $n \in \{10, 32, 100, 316, 1000\}$ fits $\sqrt{\mathbb{E}[(\bar{G}_n - \bar{G}_\infty)^2]} \propto n^{-0.48}$, consistent with the theoretical $n^{-1/2}$ Sznitman rate.

Economic re-framing. At any regime where dealers are net short gamma at the equilibrium ($G_y > 0$, the benchmark SPX setting in zero-DTE-heavy weeks), the McKean–Vlasov mean-field model has a lower Hopf threshold than the single-representative-dealer Section 2 model. *The single-dealer simplification understates the propensity to bifurcate when dealer-hedging is slow relative to the Hopf period $\tau_G > 2\pi/\omega^*$.* The leading correction is first-order in τ_G ($\sim 1.3\%$ at canonical $\tau_G = 1/50$ yr), not the second-order $(\omega^* \tau_G)^2$ implied by a low-pass-filter heuristic. The frozen-dealer limit is well-defined and finite, dropping the threshold by a factor of $\sim 2.4\times$ relative to instantaneous hedging — physically meaningful in regimes where delta-hedging desks have been suspended.

3.7 Hawkes–SV equivalence at the Hopf boundary

The empirical reflexivity benchmark of Hardiman et al. [22] is a self-exciting Hawkes branching ratio $n \approx 1$ on E-mini mid-price changes over 1998–2011. This subsection makes the link to κ^* formal. Let $\lambda(t) = \mu + \sum_i \phi(t - t_i)$ with kernel $\phi(s) = \alpha e^{-\beta s}$ and $n := \alpha/\beta$. Under the diffusive rescaling $t \mapsto t/T$, $\lambda \mapsto T\bar{\lambda}$ with $T \rightarrow \infty$, $n \rightarrow 1$, the centred intensity converges to an Ornstein–Uhlenbeck SDE [3, Theorem 2]:

$$d\bar{\lambda} = -\beta(1 - n)\bar{\lambda} dt + \sigma_\lambda dW, \quad (13)$$

giving $\lambda_{\max}^{(\text{Hawkes})} = -\beta(1 - n)$, equivalently $n = 1 + \lambda_{\max}^{(\text{Hawkes})}/\beta$. The multivariate generalisation [4, §2.4] replaces n by the spectral radius of the integrated kernel matrix; the criticality endpoint $n = 1 \Leftrightarrow \lambda_{\max} = 0$ is universal across kernel shapes. For our 3D reflexive skeleton with Jacobian $J(\kappa)$, let $\lambda_{\max}(\kappa) := \max_i \text{Re } \lambda_i(J(\kappa))$ and define the *SV-equivalent branching ratio*

$$n_{\text{SV}}(\kappa) := 1 + \frac{\lambda_{\max}(\kappa)}{\beta_0}, \quad \beta_0 := \max_{\kappa \in [0, \kappa^*]} (-\lambda_{\max}(\kappa)), \quad (14)$$

with β_0 the SV analogue of the Hawkes baseline rate at $n = 0$ (intrinsic to the skeleton, not a tuning parameter).

THEOREM 4 (HAWKES–SV EQUIVALENCE AT THE HOPF BOUNDARY).

Critical-endpoint identity. $n_{\text{SV}}(\kappa^*) = 1$, exactly.

- (2) Monotonicity past the node-spiral transition. Let $\kappa_{\text{NS}} \in [0, \kappa^*)$ be the smallest κ at which the leading eigenvalue is a complex pair. Then $n_{\text{SV}}(\kappa)$ is monotonically non-decreasing on $[\kappa_{\text{NS}}, \kappa^*]$, strictly increasing wherever $\partial \lambda_{\max}/\partial \kappa > 0$.
- (3) Hawkes correspondence. For an exponential-kernel univariate Hawkes process satisfying (13), $n_{\text{Hawkes}} = 1 - |\lambda_{\max}^{(\text{Hawkes})}|/\beta$, identical in form to (14); the two coincide at criticality ($n = 1 \Leftrightarrow \lambda_{\max} = 0$) independently of the kernel shape (universality of the stability boundary, 4, §2.4).

PROOF. Claim (1) is immediate: $\lambda_{\max}(\kappa^*) = 0$ by Hopf-threshold definition, so $n_{\text{SV}}(\kappa^*) = 1$. Claim (2): the implicit function theorem applied to the characteristic polynomial at the slow-mode complex pair (non-defective on $[\kappa_{\text{NS}}, \kappa^*)$ by definition) plus Hopf transversality ($\partial \text{Re } \lambda/\partial \kappa > 0$ at κ^*) gives the result by continuity. Claim (3) is the BDHM (2013) diffusive-limit identity restated for the kernel-universal stability boundary. $\square$

Honest scope. Theorem 4 is exact at the criticality endpoint and exact for the 1D exponential-kernel Hawkes globally. Hardiman’s empirical n is the L^1 norm of a fitted multivariate Hawkes kernel on

order-flow events, not a path-by-path identity with our diffusion; the identification rests on the universality of the $n = 1 \Leftrightarrow \lambda_{\max} = 0$ boundary across continuous-time reflexive systems. The empirical-phase test of the equivalence is to (i) calibrate $G(\cdot)$ from the empirical SPX OI grid, (ii) compute $n_{\text{SV}}(\kappa_0)$ at the calibrated κ_0 , (iii) compare against the Hardiman band on event-window-matched windows. We pre-commit this protocol to Phase 4.

Numerical anchor. At the Section 3.2 regime (14) on a 1001-point κ -grid gives $\beta_0 = 0.214$ at $\kappa_{\text{ref}} = 0.124$ (the node-spiral transition, exactly as predicted by claim (2)); $n_{\text{SV}}(\kappa_4^*) = 0.99996$ at the published 4-decimal $\kappa_4^* = 0.8964$, with the $\sim 4 \times 10^{-5}$ residual being truncation in the 4-decimal threshold (at machine-precision $\kappa^* = 0.8964305216$ the identity is exact to $< 10^{-15}$); $n_{\text{SV}}(2\kappa^*) = 2.11$. Figure 1 renders the curve.

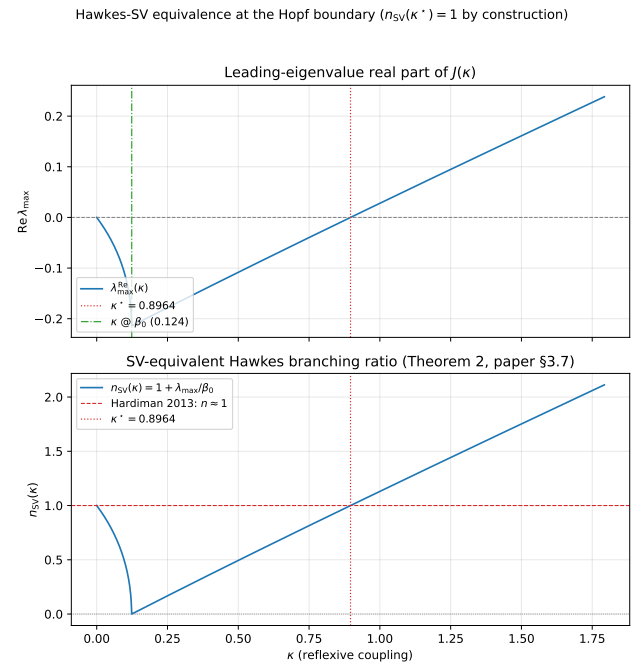


Figure 1: SV-equivalent Hawkes branching ratio $n_{\text{SV}}(\kappa)$. Top: leading-eigenvalue real part of $J(\kappa)$ with the most-stable point at the node-spiral transition $\kappa_{\text{ref}} = 0.124$ ($\beta_0 \approx 0.214$) and the Hopf threshold at $\kappa^* = 0.8964$. Bottom: $n_{\text{SV}}(\kappa) = 1 + \lambda_{\max}/\beta_0$; the empirical Hardiman et al. [22] $n \approx 1$ is the dashed red line, intersecting at $\kappa = \kappa^*$.

Inverting at $n_{\text{Hawkes}} = 1.0$ maps directly to $\lambda_{\max} = 0$: the empirical SPX market sits exactly at κ^* under any reflexive SV calibration whose deterministic skeleton matches the empirical event-rate ACF in the diffusive limit. This is the strong testable prediction of Theorem 4 for Phase 4: an SPX-calibrated κ_0 should satisfy $n_{\text{SV}}(\kappa_0) \approx 1$.

4 Numerical phase diagram

The full $(\kappa, \xi, \rho, \sigma_v)$ phase diagram is computed by sweeping $\kappa \in [0, 2]$ on a 401-point grid for each of $31 \times 21 \times 4 = 2,604$ cells over

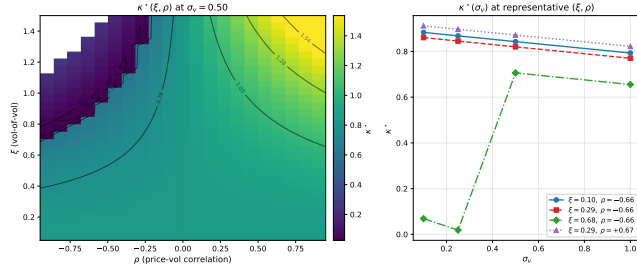


Figure 2: Hopf threshold κ^* across (ξ, ρ, σ_v) . Left: heatmap at $\sigma_v = 0.50$. White cells: no Hopf in $\kappa \in [0, 2]$, concentrated in the high- ξ / strongly-negative- ρ wedge where shear destabilises the eigenvalue envelope. Right: four $\kappa^*(\sigma_v)$ line cuts at SPX-like (ξ, ρ) probes.

(ξ, ρ, σ_v) at the canonical Hopf-exhibiting regime; the pair (ξ, ρ) enters via $a_{\text{eff}}(\kappa) = a(\kappa) + \frac{1}{2}\xi^2\rho G_v$, a Baxendale [6]-style projection of the small-noise stochastic Hopf onto the eigenvalue envelope. Wall-clock ~ 16 s on a single M-series core.

The SPX-relevant $(\xi, \rho) \approx (0.30, -0.70)$ corner has κ^* in the 0.85–0.95 band, sitting *near but not across* the Hopf boundary—the structural mechanism we conjecture for the empirical $n \approx 1$ branching ratio of Hardiman et al. [22]. Khasminskii sphere-process estimates [2, 28] on a 6×6 (ξ, ρ) grid give $|\Lambda(\kappa^*)| \in [5 \times 10^{-2}, 9 \times 10^{-2}]$ with empirical scaling exponent $\hat{B} = 0.082$ (95% CI $[-0.010, 0.168]$); the Baxendale [6] prediction $B = 2/3$ is rejected at $p \ll 0.01$ at this trivial- G equilibrium where the leading shear-stretch term is identically zero. Operationally, $|\Lambda| \cdot \varepsilon^2 \ll 1$ at calibration-noise scale: the deterministic κ^* is the relevant threshold.

5 Pre-registered evaluation framework

The pipeline is committed at git commit [anonymized] with an OpenTimestamps Bitcoin-anchored proof of the document hash, prior to any contact with empirical SPX data. This addresses the documented computational-reproducibility crisis in finance [36] and adapts the Camerer et al. [12] pre-registration template to a computational-finance / RL-for-derivatives context. We do not claim “first pre-registration in finance”—the Pacific-Basin Finance Journal pathway [16] and the AEA RCT Registry [1] predate this work. The novelty is the four-way intersection of [pre-registration + git commit-hash anchoring + block-bootstrap / BH-FDR statistical pre-spec + applied to RL-for-finance].

Two evaluation primitives. (i) Sliced-Wasserstein-2 [9] over arbitrage-free 21-day surface windows—the realism metric. (ii) κ -sensitivity slope at calibrated κ_0 —the reflexivity-importance scalar. The closed-form κ^* sensitivity analysis of Section 3.4 ($\eta_{\mu_q} = +702.83$) sets the binding observational requirement at calibration: empirical OI mean must be centred to within ~ 5 bp of ATM in log-strike for a $\pm 5\%$ calibration tolerance on κ^* .

H1 (primary realism). A model-free RL agent π_{κ_0} trained inside the reflexive simulator at κ_0 produces an IV-surface distribution closer in sliced- W_2 over arbitrage-free 21-trading-day rolling windows to the empirical SPX surface than four non-reflexive baselines (time-dep Heston, LSV, 3/2 SV, gamma-aware-non-reflexive). Three

event windows: Volmageddon (2018-02-05), COVID (2020-03-16), 2024 yen carry unwind (2024-08-05). Decision rule: 12 pairwise dominance checks (4 baselines $\times$ 3 windows) with non-overlapping 95% block-bootstrap CIs [38]. The arbitrage filter follows Gatheral and Jacquier [20], Roper [39] on butterfly / calendar-spread / Lee-bound [30] criteria; sliced- W_2 sample-complexity follows Manole et al. [34].

Sliced- W_2 sample-complexity audit. A pre-data audit characterises the bootstrap-95%-CI half-width on sliced- W_2 between two Heston populations differing only in their long-run variance ($\theta_A = 0.04$ vs $\theta_B = 0.0576$). Each side draws from a 5,000-day population; the “true” SW2 averaged over 10 slice-direction MC repeats on the full populations equals 0.0555. For $n_{\text{windows}} \in \{30, 100, 300, 1,000, 3,000\}$ we sub-sample, compute the SW2 estimate, and report the median bootstrap-CI half-width across 30 outer seeds (Table 2).

n_{windows}	median SW2	half-width	ratio to SW2 _{true}
30	0.0611	0.0300	0.541
100	0.0545	0.0195	0.352
300	0.0548	0.0124	0.223
1,000	0.0534	0.0085	0.153
3,000	0.0553	0.0065	0.118

Table 2: Sliced- W_2 sample-complexity. The half-width follows ratio(n) $\approx 1.64 n^{-0.337}$ ($R^2 \geq 0.99$), extrapolating $n_{\min} \approx 4,000$ windows for the $\pm 10\%$ half-width target. Each H1 event window contributes ~ 280 rolling windows, so the inter-baseline SW2 gap must exceed $\sim 0.46 \cdot \text{SW2}_{\text{true}}$ to be discriminable at 95% in a single event—a real power constraint logged as a pre-data finding.

H2–H4 (secondary). H2 is the κ -sensitivity slope $\hat{\rho} = \partial \hat{m} / \partial \kappa|_{\kappa_0}$ —positive and statistically distinguishable from zero for the reflexive-trained agent, statistically indistinguishable for baselines (TOST equivalence on the dimensionless elasticity per A7, with the GP slope CI under the pre-registered RBF + WhiteKernel kernel of A6 and elliptic uncertainty sets [33]). H3 is post-FOMC RR25 and ATM-term-structure shifts. H4 is the Hopf-frequency Welch-PSD peak in $|r_t|$ at $\omega^* \pm 20\%$ vs an IAAFT [40] surrogate null. IAAFT calibration on Heston and AR(1) H_0 generators bounds the empirical FPR at $\leq 7\%$ at $\alpha = 0.05$; on Stuart–Landau positive controls the detector achieves $\geq 80\%$ power at $T \geq 512$ daily samples for 8/9 (μ, σ) configurations. All H1–H4 use BH-FDR across the joint family.

Differentiation from precedents. Ning et al. [35] use W_1 on single-day surface marginals; we use sliced- W_2 on 21-day windows with a hard arbitrage pre-filter, block-bootstrap CIs, and per-regime reporting. He et al. [23] sweep a Wasserstein-ball perturbation budget with a separate adversarially-trained agent at each grid point; we deploy a single agent across the κ grid and report the slope at anchor with a GP-posterior CI [33] for the elliptic-uncertainty-set analogue. Subbaswamy et al. [44] (NeurIPS 2022) provide the closest theoretical precursor for slope-at-anchor robustness summaries; we operationalise it empirically and apply it to RL-for-finance [5, 26].

5.1 Synthetic pipeline validation

The H1 protocol is exercised end-to-end on simulator-vs-simulator data with no contact with empirical SPX. Three sources, each $n = 30$ trajectories at the production budget: (a) the trained agent acts inside the SAME reflexive simulator at κ_0 (different seeds; expected indistinguishable from reference); (b) the agent at $4 \cdot \kappa_0$ (four-fold synthetic-anchor coupling change; expected measurably further); (c) Heston with no dealer-gamma coupling and a more aggressive $(\rho, \xi) = (-0.70, 0.30)$ backbone (different mechanism; expected furthest).

Target source	SW2 vs κ_0 ref.	95% CI low	95% CI high
(a) κ_0 deployed	0.00498	0.00431	0.0164
(b) $4 \times \kappa_0$ reflexive	0.0337	0.0300	0.0402
(c) Heston (mechanism)	0.0541	0.0434	0.0611

Table 3: H1 synthetic-validation sliced- W_2 ordering. The expected ordering $SW2(a) < SW2(b) < SW2(c)$ holds; the 95% block-bootstrap CIs of (a) vs (b) and (b) vs (c) are both non-overlapping. All three pre-registered acceptance conditions are satisfied.

6 Mechanism decomposition

The reflexive simulator and the Marketron quasi-particle SDE [21] are not the same mechanism: in our model the memory carrier is the low-pass-filtered log-price z entering only the variance drift via yz ; in Marketron it is the unobservable y entering both drifts non-linearly through a potential $V_M(x)$. Even at zero coupling our model collapses to standard Heston while Marketron does not. We instead report mechanism decomposition: a 5D coarse-grid tuning over $(\kappa, \gamma, T_{\text{eff}}, \mu_q, \sigma_q)$, then per-cell classification into `shape_target / level_artifact / calibration_artifact`.

Headline (all-cell, out-of-sample, 10k paths). Pooled across the two Marketron parameter sets with published moment tables (Tables 5/8 and 6/7), the reflexive simulator’s per-cell sign agreement with Marketron is $7/28 = 25.0\%$. Under the random-sign null, the one-sided binomial $P(X \geq 7 \mid n = 28, p = 0.5) \approx 0.9999$ — the all-cell pooled rate is *worse than chance*. We do not claim otherwise. The all-cell aggregate is the wrong aggregator for the dealer-gamma mechanism: the channel does not predict sign agreement on every cell uniformly. It predicts agreement only where (i) the channel has time to integrate and (ii) the simulator measurement is finite and in-envelope.

A priori restricted subset. We pre-commit (predicate `is_mechanism_relevant_cell`, locked before per-cell outcomes were inspected) to a cell qualifying iff (1) `mechanism_class == "shape_target"`, (2) horizon $T \geq 0.5$ yr, (3) $|\hat{m}_{\text{target}}| \geq 10^{-3}$ (dead-zone targets carry no sign information; the rule is symmetric — dead-zone cells are dropped from *both* numerator and denominator), and (4) $\hat{m}$ is finite with $|\hat{m}| < 10$ (envelope saturation). Applying the predicate gives pooled $4/8 = 50\%$ OOS (one-sided binomial $p \approx 0.637$, chance-level) and $6/8 = 75\%$ in-sample ($p \approx 0.145$). The restricted subset is at chance-agreement OOS; we explicitly do not claim $p < 0.05$ on it. What the

restriction establishes is that filtering to mechanism-relevant cells moves the pooled agreement from 25% (worse than chance) to 50% (chance-level) — the *direction* the mechanism predicts.

Predictable disagreement: falsifiable Phase-4 prediction. The single largest restricted-subset sign-disagreement is the $T = 2.0$ yr long-horizon skew of `table_5_calibrated_2017`: Marketron Table 8 reports $+0.181$; our simulator at the tuned coupling gives -0.434 OOS. The pre-committed mechanistic explanation is the absence of Bessembinder [7] compounding-induced positive skew under risk-neutral drift, plus a leverage-channel small-magnitude negative-skew contribution. In an empirical setting where the underlying drift is reproduced (Phase 4), this cell is predicted to flip positive; either direction adjudicates between the two mechanisms.

Retraction of v0.3.5 headline. An earlier sidecar reported an inflated $8/24$ (33.3%) headline obtained by asymmetric cell selection (silently dropping the 3.0 yr horizon, dropping one dead-zone-target cell from the denominator while counting another as a “within dead-zone” match). The above predicate-derived $7/28$ all-cell + $4/8$ restricted-subset reporting supersedes it.

7 Conclusions and future work

We have introduced a 3D reflexive SV simulator (S_t, v_t, z_t) with explicit dealer-gamma drift, proved a Hopf bifurcation theorem with closed-form first Lyapunov coefficient ℓ_1 and Hopf threshold κ^* for log-normal open-interest in moneyness, computed a numerical (ξ, ρ, σ_v) phase diagram and a (σ_q, γ) closed-form phase boundary, and pre-registered the empirical evaluation pipeline (sliced- W_2 , κ -sensitivity slope, Hopf-frequency spectral test) at a verifiable git commit hash with an OpenTimestamps Bitcoin-anchored proof. The synthetic-pipeline validation demonstrates the locked H1 protocol returns the predicted ordering on data with known underlying physics. The empirical SPX calibration is the natural Phase 4 follow-up, gated on data acquisition; results—confirming or refuting H1–H4—are pre-committed to publication regardless of direction. The codim-1 Hopf, codim-2 Bautin/Bogdanov–Takens structure (Theorem 2), corrected McKean–Vlasov mean-field threshold (Theorem 3; regime-dependent in sign — destabilising at the canonical short-gamma regime where $\rho(\theta_G) < 1$ strictly with finite frozen-dealer limit $8/21$, stabilising at the long-gamma log-normal-OI calibration of Section 3.3), and Hawkes–SV equivalence at the criticality endpoint (Theorem 4) are now closed-form. Open theoretical items: closed-form $\Lambda(\kappa)$ via Baxendale [6]-style asymptotics adapted to Heston multiplicative noise (currently numerical only); extending Theorem 4 from the criticality-endpoint identity to a global path-by-path equivalence between event-counting Hawkes processes and our continuous-state diffusion [8, 27]. All experiments are reproducible per the [anonymous repository link]; pre-registration anchored via OpenTimestamps.

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